```java
import java.util.*;
class Stack
{
    int s[];
    int size,top;

    Stack(int n)
    {
        size=n;
        s=new int[size];
        top=-1;
    }
    void push(int n)
    {
        if(top==size-1)
        {
            System.out.println("STACK OVERFLOW");

        }
        else
        {
            top=top+1;
            s[top]=n;
        }
    }

    int pop()
    {
        if(top==-1)
        {
            System.out.println("STACK UNDERFLOW");
            return -999;
        }
        else
        {
            int val=s[top];
            top-=1;
            return val;
        }
    }

    void display()
    {
        if(top ==-1)
            System.out.println("The stack is empty");

        else
        {
            System.out.println("The elements in the stack are as follows");
            for(int i= top;i>=0;i--)
```

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                System.out.println(s[i]);
            }
        }

        public static void main()
        {
            Scanner sc=new Scanner(System.in);
            System.out.println("ENTER THE SIZE OF STACK");
            int sz=sc.nextInt();
            Stack ob=new Stack(sz);
            while(true)
            {
                System.out.println("ENTER 'I' if you wish to input data ");
                System.out.println("ENTER 'R' if you wish to remove data ");
                System.out.println("ENTER 'D' if you wish to display data ");
                System.out.println("ENTER '0' if you wish to termiate");
                char ch=sc.next().charAt(0);
                switch(ch)
                {
                    case 'I':
                    case 'i':System.out.println("ENTER DATA TO INSERT:");
                             ob.push(sc.nextInt());
                             break;
                    case 'R':
                    case 'r':int y=ob.pop();
                             if(y!=-999)
                              System.out.println("Removed  "+y);
                             break;

                    case 'D':
                    case 'd':ob.display();
                             break;

                    case '0':System.out.println("NOW TERMINATING");
                             System.exit(0);

                    default :System.out.println("INVALID INPUT !!");
                }
                }
            }
        }
```